



REVERSE-ENGINEERING A PUBLISHED IMMERSION COOLING BENCHMARK

THERMAL SIMULATION · SOLIDWORKS FLOW SIMULATION · CFD

TOOLS & METHODS:

SOLIDWORKS reverse-engineering from published imagery · SOLIDWORKS Flow Simulation steady-state conjugate heat transfer · k - ϵ turbulence with turbulent boundary-layer treatment · mass-flow inlet boundary conditions · distributed volumetric heat sources across four die regions · propeller geometry sized from the heat sink envelope · local mesh refinement on the die, propellers and heat sink at 7 cells across channels · parametric flow sweep at two power levels · goal-based convergence on temperature and heat transfer rate.

EXECUTIVE SUMMARY:

Single-phase immersion cooling is moving past the 1000 W threshold, and a published Intel and Submer case study set the reference point: a forced-convection heat sink holding a case-to-fluid thermal resistance near 0.044 °C/W across a 100 W to 1000 W sweep. Any company developing a competing thermal design has to answer one question before committing to hardware. How does our concept compare with that?

Building the benchmark to measure against is expensive. This programme built it in simulation instead. The heat sink and its propellers were reconstructed in CAD from published photographs and performance figures, the flow tuned to the published specification, and 800 W and 1000 W applied across four die regions of the processor the benchmark used. Twelve converged runs across a fivefold flow sweep reproduce the benchmark's governing behaviour and predict case-level thermal resistance within 14% of measurement, which is close enough that a new design evaluated in the same model can be compared with confidence.

KEY RESULTS:

- **Case-to-inlet resistance of 0.0369 to 0.0385 °C/W**, moving only 4.3% across a fivefold increase in bulk flow.
- **Reproduces the benchmark's own headline finding** that server-level flow rate barely affects case temperature: 1.6 °C across the full sweep.
- **Fin heat transfer coefficient rises 10%** over the same sweep while system resistance barely moves, placing the bottleneck in conduction rather than convection.
- **Thermal resistance is constant with power:** temperature rise scales 30.81 to 38.65 K for a 1.25x load step, a ratio of 1.254.

- **Offset quantified and explained:** the model predicts the case 5 to 6 °C cooler than measurement, consistent with omitted interface resistance.
- **Twelve converged steady-state runs** at 800 W and 1000 W across 1 to 5 LPM.

INTRODUCTION:

Rising thermal design power is pushing data centres from air to liquid, and single-phase immersion is one of the leading routes because it insulates the hardware, cuts infrastructure and scales. Its limit has historically been heat flux: natural convection in a dielectric bath handles roughly 400 W before the fluid stops carrying heat away fast enough. A published Intel and Submer case study addressed that by adding propellers to an immersion-optimised heat sink, reporting case-to-fluid resistance near 0.044 °C/W sustained from 100 W to 1000 W, competitive with water-based direct liquid cooling.

That result is the number a new entrant has to beat, and it creates a problem. The benchmark existed as a published document: photographs, performance curves and stated test conditions, with no drawings, no model and no hardware to hand. Buying and instrumenting a competitor's heat sink is slow and expensive. The question this work set out to answer is whether a simulation baseline built from published material alone can be made accurate enough that a design under development can be compared against it credibly.

TEST SETUP:



FIG 1: The computational domain at 4 LPM, viewed from above. The reconstructed fin array sits in the dielectric bath alongside the other modules in the tank; probe values run from 327.8 K at the fin base to 319.0 K at the tip.

Reconstructing the geometry. The heat sink body, fin pitch and overall envelope were rebuilt in SOLIDWORKS from the published imagery and the stated physical dimensions. The propellers had no published geometry at all, so they were sized from the heat sink envelope they mount to, and the resulting flow was then tuned until the model matched the flow specification quoted for the same processor. This is the step that makes or breaks a reverse-engineered baseline: the geometry only has to be right enough that the flow field and the conduction path behave correctly, and the way to confirm that is to match a published operating point rather than to guess at dimensions.

Boundary conditions. Flow enters through mass-flow inlets rather than a prescribed velocity, which keeps the delivered flow rate fixed as fluid properties change with temperature. The 4 LPM case corresponds to 0.0547 kg/s at the dielectric's density. Inlet temperature was held at 40 °C, matching the published test condition, and the bath was modelled as conjugate heat transfer so that conduction through the die stack, spreader and fins is solved together with the fluid.

Heat application. Rather than applying a single lumped load, 800 W and 1000 W were distributed across the four main die regions of the processor, matching its multi-tile layout. That matters for where the peak lands: a single uniform source over the whole package spreads heat more evenly than the real silicon does and flatters the result.

The sweep. Bulk flow was varied from 1 to 5 LPM at both power levels, twelve runs in total. Each was driven to steady state against goals on maximum temperature, heat sink temperature and heat transfer rate. Turbulence used the k- ϵ model with a turbulent boundary-layer treatment, and local mesh refinement was applied to the die, both propellers and the heat sink, resolving seven cells across the fin channels.

Global heat transfer out of the domain settled within 2% of the applied load, confirming the energy balance closed.

One honest caveat on accuracy. Agreement is quoted here as thermal resistance, which is the quantity the model actually predicts. Quoting agreement on absolute temperature instead would read 93%, but roughly 40 of those degrees are the inlet condition imposed as a boundary condition rather than anything the solver computed. Resistance is the harder and more meaningful test.

RESULTS:

FIG 2: Steady-state section through the package at 4 LPM. Heat spreads from the die through the spreader and base into the fin block, with the dielectric bath at 317 K.

Where the temperature actually drops. The section in Figure 2 shows the gradient concentrated in the die stack and the spreader, not in the fluid. The top-down view reinforces it: probe values fall from 327.8 K at the fin base to 319.0 K at the tip, an 8.8 K drop along the fin itself. A fin carrying that much gradient is not convection-starved at its surface, it is conduction-limited along its length, and the same is true of the path feeding it.

FIG 3: Case-to-inlet thermal resistance against bulk flow rate at both power levels, with the published measurement band shaded.

Flow rate barely matters, and that is the benchmark's own finding. Case-to-inlet resistance sits between 0.0369 and 0.0385 °C/W and moves 4.3% across a fivefold flow increase. Case temperature changes by 1.6 °C over the same range. The published study reported the same insensitivity from its own measurements, noting that fluid takes the path of least resistance and largely avoids travelling between the fins. Reproducing that behaviour independently, from a model built



without access to the hardware, is the strongest evidence that the reconstruction captured the real flow physics.

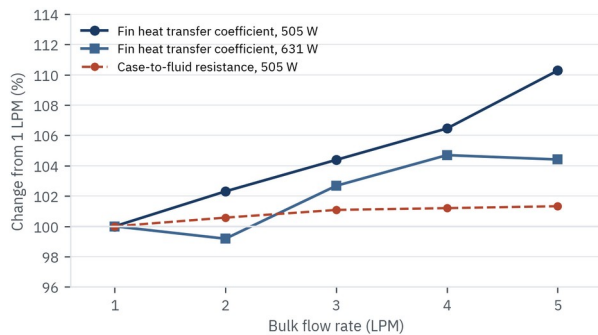


FIG 4: Fin heat transfer coefficient and system thermal resistance, both normalised to their 1 LPM values. Convection improves while the system does not.

The mechanism behind it. Fin heat transfer coefficient climbs about 10% from 1 to 5 LPM while system thermal resistance changes by 4.3%. Convection at the fin surface is genuinely improving and the device barely responds, because the limiting resistance is not at the fins. It is in the conduction path through the die stack, the interface material and the heat sink base. This is the quantitative version of the published study's qualitative observation, and it is the practical design lesson: at this power density, effort spent driving more fluid through the fins returns very little, while effort spent on the interface stack returns a great deal.

Resistance is constant with power. Temperature rise scales from 30.81 K at 800 W to 38.65 K at 1000 W, a ratio of 1.254 against a load ratio of 1.25. Thermal resistance is therefore effectively independent of load across this range, which is what allows the model to be used as a reference at power levels it was not explicitly run at.

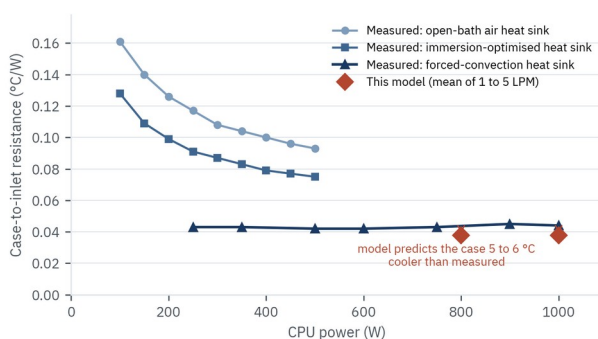


FIG 5: Simulated resistance against the three configurations measured in the published study. The model tracks the forced-convection benchmark with a consistent optimistic offset.

WHERE THE MODEL AND THE MEASUREMENT DISAGREE

Simulated case-to-inlet resistance averages 0.0377 °C/W against roughly 0.0437 °C/W measured, so the model predicts the case 5 to 6 °C cooler than the hardware, an agreement of 86%. The missing resistance is about 0.0060 °C/W. Bulk indium foil at the published thickness accounts for

under 0.001 °C/W, so the foil itself is not the cause. Split across the die-to-spreader and spreader-to-base joints, the shortfall corresponds to a contact conductance near 100,000 W/m²K per interface, which sits at the optimistic end of the normal range for a metallic joint. Contact resistance omitted at those interfaces plausibly accounts for the whole gap, and adding it is the next run.

Why the offset does not invalidate the baseline. A systematic offset arising from a shared modelling assumption largely cancels when two designs are compared inside the same model. The absolute number should be quoted with its 14% caveat, but the difference between the benchmark and a new concept evaluated identically is considerably more reliable than either figure alone. What the model demonstrably gets right is the behaviour: the flow insensitivity, the convection-to-conduction decoupling, and the linear scaling with load.

APPLICATIONS:

The immediate use is comparative. A thermal design under development can be dropped into the same domain, meshed the same way and run at the same boundary conditions, and its thermal resistance read against a benchmark that would otherwise have required buying, instrumenting and testing a competitor's hardware. The model also answers questions the published study does not: it exposes the temperature field inside the die stack, shows where the conduction bottleneck sits, and allows the flow rate, fluid and fin geometry to be varied at no marginal cost.

The transferable method is reconstruction and behavioural validation. When a competitor publishes performance data but no geometry, a model can still be anchored by matching a published operating point and then checking that it reproduces the trends the publication reports. Behavioural agreement is a weaker claim than absolute agreement, and it should be stated as such, but it is frequently enough to support a design decision, and it is available in days rather than the months a physical benchmark would take.

DISCUSS THIS WORK:

This paper summarises one project. If the same problem appears in your programme, I am happy to walk through the method, the data behind it, and how it would map onto your hardware. emincherif@gmail.com